🔢 Arithmetic Fundamentals: The Complete Foundation

Arithmetic Fundamentals Study Guide

📚 Prerequisite knowledge for Arithmetic Reasoning | Master the fundamentals!

> 💡 **Purpose:** This guide explains the "why" and "how" behind arithmetic operations. Master these fundamentals before tackling word problems!

📖 Overview

This foundational guide covers the core arithmetic concepts you need to master for Arithmetic Reasoning:

- **Basic Operations** - Addition, subtraction, multiplication, division
- **Fractions** - Common denominators, operations, conversions
- **Decimals & Percentages** - Conversions, calculations
- **Unit Conversions** - Systematic approach to converting units
- **Ratios & Proportions** - Understanding relationships
- **When to Multiply vs Divide** - Decision framework
- **Mental Math** - Strategies for speed and accuracy

1️⃣ Basic Operations Foundation

Addition: Combining Quantities

> 🟢 What Addition Means
>
> **Addition** = Combining two or more quantities to find the total
>
> Think: "How many in all?" or "What's the total?"

Visual Example:

```
3 apples + 4 apples = 7 apples total
```

Key Words: sum, total, combined, together, plus, add

Example 1: Simple Addition

> **Problem**
>
> Sarah has 23 books. She buys 17 more books. How many books does she have in total?

Step-by-Step Solution:

Step 1: Identify what you're adding

- Sarah's current books: 23
- New books: 17
- We want the total

Step 2: Set up the addition

```
  23
+ 17
----
```

Step 3: Add the ones place (right column)

- 3 + 7 = 10
- Write 0, carry 1 to tens place

Step 4: Add the tens place (left column)

- 2 + 1 + 1 (carried) = 4
- Write 4

Step 5: Write the answer

```
  23
+ 17
----
  40
```

Answer: 40 books

Why This Works: Addition combines quantities. We add the ones place first, then the tens place, carrying when needed.

Example 2: Addition with Carrying

> **Problem**

> A store sold $156 worth of merchandise in the morning and $287 in the afternoon. What was the total sales for the day?

Step-by-Step Solution:

Step 1: Identify the numbers to add

- Morning sales: $156
- Afternoon sales: $287
- Total: ?

Step 2: Align numbers by place value

```
  156
+ 287
-----
```

Step 3: Add ones place

- 6 + 7 = 13
- Write 3, carry 1

Step 4: Add tens place

- 5 + 8 + 1 (carried) = 14
- Write 4, carry 1

Step 5: Add hundreds place

- 1 + 2 + 1 (carried) = 4
- Write 4

Step 6: Final answer

```
  156
+ 287
-----
  443
```

Answer: $443

Why This Works: When a column adds up to 10 or more, we "carry" the extra to the next column, just like counting past 9.

Example 3: Adding Multiple Numbers

Problem

Jenny's test scores are 93, 89, 96, and 98. What is the sum of all her test scores?

Step-by-Step Solution:

Step 1: List all numbers to add

- 93 + 89 + 96 + 98

Step 2: Add two at a time (easier mentally)

- First pair: 93 + 89 = 182
- Second pair: 96 + 98 = 194

Step 3: Add the results

- 182 + 194 = 376

OR add step-by-step:

```
  93
  89
  96
+ 98
-----
```

Step 4: Add column by column

- Ones: 3 + 9 + 6 + 8 = 26 → Write 6, carry 2
- Tens: 9 + 8 + 9 + 9 + 2 (carried) = 37 → Write 7, carry 3
- Hundreds: 3 (carried) = 3

Answer: 376

Why This Works: You can add numbers in any order. Grouping makes it easier to manage carrying.

Practice Problems: Addition

Try These Yourself

1. 45 + 38 = ?
2. 127 + 89 = ?
3. 234 + 156 + 78 = ?
4. A car traveled 156 miles on Monday and 287 miles on Tuesday. How many miles total?

> 5. Tom spent $23.50 on groceries and $17.75 on gas. What was his total spending?

Answer Key: Addition

Solutions

1. 45 + 38 = ?

Solution:

Ones: 5 + 8 = 13 → Write 3, carry 1

Tens: 4 + 3 + 1 = 8

Answer: 83

2. 127 + 89 = ?

Solution:

Ones: 7 + 9 = 16 → Write 6, carry 1

Tens: 2 + 8 + 1 = 11 → Write 1, carry 1

Hundreds: 1 + 1 = 2

Answer: 216

3. 234 + 156 + 78 = ?

Solution:

Add first two: 234 + 156 = 390

Add third: 390 + 78 = 468

Answer: 468

4. A car traveled 156 miles on Monday and 287 miles on Tuesday. How many miles total?

Solution:

156 + 287 = 443

Answer: 443 miles

5. Tom spent $23.50 on groceries and $17.75 on gas. What was his total spending?

Solution:

Align decimals: $23.50 + $17.75

Hundredths: 0 + 5 = 5

Tenths: 5 + 7 = 12 → Write 2, carry 1

Ones: 3 + 7 + 1 = 11 → Write 1, carry 1

Tens: 2 + 1 + 1 = 4

Answer: $41.25

Subtraction: Taking Away or Finding Difference

🔴 What Subtraction Means

Subtraction = Taking away from a quantity OR finding the difference between two quantities

Think: "How many are left?" or "How much more/less?"

Visual Example:

```
10 cookies - 3 cookies = 7 cookies left
OR
10 cookies - 3 cookies = 7 cookies difference
```

Key Words: difference, less, minus, subtract, take away, remaining

---

Example 1: Simple Subtraction

Problem

Miriam bought 5 cases of motor oil. A case normally costs $24.00, but she paid $22.50 per case. How much did she save per case?

Step-by-Step Solution:

Step 1: Identify what you're finding

- Regular price: $24.00
- Sale price: $22.50
- We want the difference (savings)

Step 2: Set up subtraction

```
  24.00
- 22.50
-------
```

Step 3: Subtract hundredths place

- 0 - 0 = 0

Step 4: Subtract tenths place

- 0 - 5 → Can't do directly, need to borrow
- Borrow from ones: 4 becomes 3, 0 becomes 10
- 10 - 5 = 5

Step 5: Subtract ones place

- 3 - 2 = 1

Step 6: Subtract tens place

- 2 - 2 = 0

Answer: $1.50 saved per case

Why This Works: When you can't subtract in a column, borrow from the next higher place value.

Example 2: Subtraction with Borrowing

> **Problem**
>
> A security guard walks 6 city blocks. If he walks at a pace of 8 blocks every 30 minutes, how long does it take to walk one block?

Step-by-Step Solution:

Step 1: Understand the problem

- Total blocks: 6
- Time for 8 blocks: 30 minutes
- We need time per block

Step 2: This is actually division (30 ÷ 8), but let's think of it as subtraction

- How many times can we subtract 8 from 30?
- Actually, this is better solved with division (see Division section)

Alternative approach - Finding difference:

- If 8 blocks take 30 minutes

- Time per block = 30 ÷ 8 = 3.75 minutes

Step 3: Using subtraction to verify:

- 30 - 8 = 22 (1 block walked, 22 min left)
- 22 - 8 = 14 (2 blocks, 14 min left)
- 14 - 8 = 6 (3 blocks, 6 min left)
- 6 < 8, so we have 3 full blocks + 6/8 = 3.75 minutes per block

Answer: 3.75 minutes per block

Why This Works: Subtraction can help us understand division by seeing how many times we can subtract.

Example 3: Finding Difference

> **Problem**
>
> An office building has 30 employees with 42 square feet per employee. If 5 more employees are hired, each employee will have 36 square feet. How much less space is that?

Step-by-Step Solution:

Step 1: Identify the two amounts

- Original space: 42 sq ft per person
- New space: 36 sq ft per person
- Find the difference

Step 2: Set up subtraction

```
  42
- 36
----
```

Step 3: Subtract ones place

- 2 - 6 → Can't do, borrow from tens
- 4 becomes 3, 2 becomes 12
- 12 - 6 = 6

Step 4: Subtract tens place

- 3 - 3 = 0

Answer: 6 square feet less per employee

Why This Works: Finding the difference tells us how much one quantity is more or less than another.

Practice Problems: Subtraction

Try These Yourself

1. 87 - 39 = ?
2. 156 - 89 = ?
3. 500 - 237 = ?
4. If a shirt costs $45.00 and is on sale for $32.50, how much do you save?
5. The temperature dropped from 78°F to 65°F. What was the temperature change?

Answer Key: Subtraction

Solutions

1. 87 - 39 = ?

Solution:

Ones: 7 - 9 → Borrow: 17 - 9 = 8

Tens: 7 - 3 = 4 (after borrowing)

Answer: 48

2. 156 - 89 = ?

Solution:

Ones: 6 - 9 → Borrow: 16 - 9 = 7

Tens: 4 - 8 → Borrow: 14 - 8 = 6

Hundreds: 0 - 0 = 0

Answer: 67

3. 500 - 237 = ?

Solution:

Ones: 0 - 7 → Borrow: 10 - 7 = 3

Tens: 9 - 3 = 6 (after borrowing)

Hundreds: 4 - 2 = 2 (after borrowing)

Answer: 263

4. If a shirt costs $45.00 and is on sale for $32.50, how much do you save?

Solution:

$45.00 - $32.50

Hundredths: 0 - 0 = 0

Tenths: 0 - 5 → Borrow: 10 - 5 = 5

Ones: 4 - 2 = 2 (after borrowing)

Tens: 4 - 3 = 1

Answer: $12.50

5. The temperature dropped from 78°F to 65°F. What was the temperature change?

Solution:

78 - 65 = 13

Answer: 13°F decrease

Multiplication: Repeated Addition

> 🔵 **What Multiplication Means**
>
> **Multiplication** = Adding the same number multiple times
>
> Think: "How many groups of X?" or "X times Y"

Visual Example:

```
3 × 4 = 4 + 4 + 4 = 12
(3 groups of 4)
```

Key Words: times, multiplied by, product, groups of, per (when combined with quantity)

When to Multiply:

- Finding total when you have "X per Y"
- Finding area (length × width)
- Finding total cost (price × quantity)
- Finding distance (speed × time)

Example 1: Basic Multiplication

Problem

If a car is towed 12 miles and the charge is $3.50 per mile, how much does the tow cost?

Step-by-Step Solution:

Step 1: Identify what you're multiplying

- Rate: $3.50 per mile
- Quantity: 12 miles
- Formula: Cost = Rate × Quantity

Step 2: Set up multiplication

```
  3.50
×   12
------
```

Step 3: Multiply by ones place (2)

- 2 × 0 = 0 (hundredths)
- 2 × 5 = 10 → Write 0, carry 1 (tenths)
- 2 × 3 + 1 = 7 (ones)
- Result: 7.00

Step 4: Multiply by tens place (1, which is actually 10)

- 1 × 0 = 0 (hundredths)
- 1 × 5 = 5 (tenths)
- 1 × 3 = 3 (ones)
- Result: 35.0 (shifted one place left because it's tens)

Step 5: Add the results

```
   7.00
+ 35.00
------
  42.00
```

Answer: $42.00

Why This Works: "Per mile" means multiply. Each mile costs $3.50, so 12 miles = 12 × $3.50.

Example 2: Multiplication with Carrying

Problem

How many square feet are needed to carpet a 12-foot × 12-foot room?

Step-by-Step Solution:

Step 1: Identify the formula

- Area = Length × Width
- Length: 12 feet
- Width: 12 feet

Step 2: Set up multiplication

```
  12
× 12
----
```

Step 3: Multiply by ones place (2)

- 2 × 2 = 4 (ones)
- 2 × 1 = 2 (tens)
- Result: 24

Step 4: Multiply by tens place (1, which is 10)

- 1 × 2 = 2 (tens)
- 1 × 1 = 1 (hundreds)
- Result: 120 (shifted left)

Step 5: Add results

```
   24
+ 120
-----
  144
```

Answer: 144 square feet

Why This Works: Area measures how many square units fit in a space. 12 × 12 = 144 square feet.

Example 3: Word Problem - "Per" Means Multiply

Problem

> A waitress earns an average tip of 12% of the cost of food. If she serves $375 worth of food, how much in tips does she earn?

Step-by-Step Solution:

Step 1: Identify the "per" relationship

- Rate: 12% per dollar of food
- Quantity: $375
- Tips = 12% × $375

Step 2: Convert percentage to decimal

- 12% = 12 ÷ 100 = 0.12

Step 3: Multiply

```
  375
× 0.12
------
```

Step 4: Multiply by hundredths (2)

- 2 × 5 = 10 → Write 0, carry 1
- 2 × 7 + 1 = 15 → Write 5, carry 1
- 2 × 3 + 1 = 7
- Result: 750 (but this is actually 7.50 because we're multiplying by 0.12)

Step 5: Multiply by tenths (1)

- 1 × 5 = 5
- 1 × 7 = 7
- 1 × 3 = 3
- Result: 37.5 (shifted one place)

Step 6: Add (or use shortcut: 375 × 0.12)

- 375 × 0.12 = 45.00

Answer: $45.00

Why This Works: "12% of" means multiply. 12% of $375 = 0.12 × 375 = 45.

Practice Problems: Multiplication

> **Try These Yourself**

1. 24 × 15 = ?
2. 8.5 × 6 = ?
3. 144 × 3 = ?
4. If you earn $12.30 per hour and work 40 hours, what's your total pay?
5. A rectangle is 16 feet long and 8 feet wide. What's its area?

Answer Key: Multiplication

Solutions

1. 24 × 15 = ?

Solution:

Method 1: 24 × 15 = 24 × (10 + 5) = 240 + 120 = 360

Method 2: Standard multiplication

24 × 5 = 120, 24 × 10 = 240, Total = 360

Answer: 360

2. 8.5 × 6 = ?

Solution:

8.5 × 6 = (8 × 6) + (0.5 × 6) = 48 + 3 = 51

Answer: 51

3. 144 × 3 = ?

Solution:

144 × 3 = (100 × 3) + (40 × 3) + (4 × 3) = 300 + 120 + 12 = 432

Answer: 432

4. If you earn $12.30 per hour and work 40 hours, what's your total pay?

Solution:

$12.30 × 40 = $492.00

Answer: $492.00

5. A rectangle is 16 feet long and 8 feet wide. What's its area?

Solution:

Area = Length × Width = 16 × 8 = 128

Answer: 128 square feet

Division: Splitting or Grouping

🟠 **What Division Means**

Division = Splitting into equal groups OR finding how many times one number fits into another

Think: "How many in each group?" or "How many groups?"

Visual Example:

```
12 ÷ 3 = 4
(12 split into 3 groups = 4 in each group)
OR
(How many groups of 3 fit into 12? = 4 groups)
```

Key Words: divided by, split, share equally, per (when finding rate), quotient

When to Divide:

- Finding average (total ÷ count)
- Finding rate (total ÷ time)
- Finding unit price (total cost ÷ quantity)
- Converting units (larger to smaller)

Example 1: Basic Division

Problem

A sales manager buys antacid in bottles by the gross. If he uses 3 bottles per day, how long will 144 bottles (1 gross) last?

Step-by-Step Solution:

Step 1: Identify what you're dividing

- Total bottles: 144 (1 gross)
- Usage rate: 3 bottles per day
- Find: How many days?

Step 2: Set up division

- Days = Total ÷ Rate per day
- Days = 144 ÷ 3

Step 3: Divide using long division

```
   48
3 ) 144
   -12
   ---
    24
   -24
   ---
     0
```

Step 4: Check your work

- 48 days × 3 bottles/day = 144 bottles ✓

Answer: 48 days

Why This Works: Division splits the total into equal groups. 144 bottles ÷ 3 per day = 48 days.

Example 2: Finding Average (Division)

> **Problem**
>
> Darla spent $120.37 in January, $108.45 in February, and $114.86 in March. What was her average monthly spending?

Step-by-Step Solution:

Step 1: Identify what you need

- Average = Sum ÷ Count
- Sum: $120.37 + $108.45 + $114.86
- Count: 3 months

Step 2: Add the three amounts first

```
  120.37
  108.45
+ 114.86
--------
  343.68
```

Step 3: Divide sum by count

- Average = $343.68 ÷ 3

Step 4: Long division

```
  114.56
3 ) 343.68
   -3
   ---
    04
   - 3
   ---
    13
   -12
   ---
     16
    -15
    ---
      18
     -18
     ---
       0
```

Answer: $114.56 average per month

Why This Works: Average divides the total equally among all items. Total spending ÷ number of months = average per month.

Example 3: Finding Rate (Division)

> **Problem**
>
> Keith drives 1,650 miles from Reno to Kansas City. If he averages 50 miles per hour, how many hours will the trip take?

Step-by-Step Solution:

Step 1: Identify the formula

- Distance = Speed × Time
- Rearrange: Time = Distance ÷ Speed

Step 2: Plug in values

- Time = 1,650 miles ÷ 50 mph

Step 3: Divide

```
     33
50 ) 1650
    -150
    ----
     150
    -150
    ----
       0
```

Step 4: Verify

- 33 hours × 50 mph = 1,650 miles ✓

Answer: 33 hours

Why This Works: When you know total distance and speed, divide to find time. Distance ÷ Speed = Time.

Practice Problems: Division

Try These Yourself

1. 144 ÷ 12 = ?
2. 375 ÷ 25 = ?
3. 1,800 ÷ 6 = ?
4. If you travel 336 miles and use 16 gallons of gas, what's your miles per gallon?
5. An office has 1,260 square feet for 35 employees. How much space per employee?

Answer Key: Division

Solutions

1. 144 ÷ 12 = ?

Solution:

How many times does 12 fit into 144?

12 × 12 = 144

Answer: 12

2. 375 ÷ 25 = ?

Solution:

25 × 15 = 375

Answer: 15

3. 1,800 ÷ 6 = ?

Solution:

1,800 ÷ 6 = 300

(Check: 300 × 6 = 1,800)

Answer: 300

4. If you travel 336 miles and use 16 gallons of gas, what's your miles per gallon?

Solution:

MPG = Distance ÷ Gallons = 336 ÷ 16 = 21

Answer: 21 miles per gallon

5. An office has 1,260 square feet for 35 employees. How much space per employee?

Solution:

Space per person = Total ÷ People = 1,260 ÷ 35

35 × 36 = 1,260

Answer: 36 square feet per employee

Order of Operations: PEMDAS

🟣 Critical Rule: PEMDAS

Parentheses → **E**xponents → **M**ultiplication/**D**ivision → **A**ddition/**S**ubtraction

Multiplication and Division are equal - do left to right

Addition and Subtraction are equal - do left to right

Example:

```
8 + 3 × 2 = ?
Wrong: 8 + 3 × 2 = 11 × 2 = 22 ❌
Right: 8 + 3 × 2 = 8 + 6 = 14 ✓
(Multiplication before addition!)
```

Step-by-step:

1. **P**arentheses: (8 + 3) × 2 = 11 × 2 = 22
2. **E**xponents: 2^3 + 4 = 8 + 4 = 12
3. **M**ultiplication/**D**ivision (left to right): 8 ÷ 2 × 4 = 4 × 4 = 16
4. **A**ddition/**S**ubtraction (left to right): 10 - 3 + 2 = 7 + 2 = 9

Memory Tip: "Please Excuse My Dear Aunt Sally"

2️⃣ Fractions: The Complete Guide

Understanding Fractions

🟢 What Fractions Represent

A fraction shows **parts of a whole**

Numerator (top) = How many parts you have

Denominator (bottom) = How many equal parts the whole is divided into

Visual Model:

```
¾ means: 3 out of 4 equal parts
[▓▓▓▓] [▓▓▓▓] [▓▓▓▓] [    ]
  ↑        ↑        ↑       ↑
 3 parts shaded, 1 part unshaded = ¾ of the whole
```

Finding Common Denominators: Step-by-Step

🔴 Why We Need Common Denominators

You can only add or subtract fractions when they have the **same denominator**

Think: You can't add apples and oranges unless you convert them to "fruit"!

Method 1: List Multiples (Easiest)

Example 1: Simple Fractions (¼ + ⅓)

Problem

Find the common denominator for ¼ and ⅓

Step-by-Step Solution:

Step 1: List multiples of each denominator

- Multiples of 4: 4, 8, **12**, 16, 20, 24...
- Multiples of 3: 3, 6, 9, **12**, 15, 18...

Step 2: Find the smallest number in both lists

- **LCD = 12** (Least Common Denominator)

Step 3: Convert $\frac{1}{4}$ to have denominator 12

- Ask: "What times 4 equals 12?" → 3
- Multiply numerator and denominator by 3: $\frac{1}{4} \times \frac{3}{3} = \frac{3}{12}$
- Check: 1 × 3 = 3, 4 × 3 = 12 ✓

Step 4: Convert $\frac{1}{3}$ to have denominator 12

- Ask: "What times 3 equals 12?" → 4
- Multiply numerator and denominator by 4: $\frac{1}{3} \times \frac{4}{4} = \frac{4}{12}$
- Check: 1 × 4 = 4, 3 × 4 = 12 ✓

Step 5: Now both fractions have the same denominator

- $\frac{1}{4} = \frac{3}{12}$
- $\frac{1}{3} = \frac{4}{12}$
- **Now you can add:** $\frac{3}{12} + \frac{4}{12} = \frac{7}{12}$

Why This Works: Multiplying by $\frac{3}{3}$ or $\frac{4}{4}$ is the same as multiplying by 1, so the value doesn't change. We're just cutting the pieces into smaller, equal-sized parts.

Example 2: More Complex ($\frac{1}{6} + \frac{8}{15}$)

> **Problem**
>
> Find the common denominator for $\frac{1}{6}$ and $\frac{8}{15}$

Step-by-Step Solution:

Step 1: List multiples (this might take longer, so use prime factorization instead)

- Multiples of 6: 6, 12, 18, 24, **30**, 36...
- Multiples of 15: 15, **30**, 45, 60...
- LCD = 30

Step 2: Convert $\frac{1}{6}$ to have denominator 30

- Ask: "What times 6 equals 30?" → 5
- Multiply: $\frac{1}{6} \times \frac{5}{5} = \frac{5}{30}$

Step 3: Convert $^{8}/_{15}$ to have denominator 30

- Ask: "What times 15 equals 30?" → 2
- Multiply: $^{8}/_{15} \times {}^{2}/_{2} = {}^{16}/_{30}$

Step 4: Both fractions now have denominator 30

- $^{1}/_{6} = {}^{5}/_{30}$
- $^{8}/_{15} = {}^{16}/_{30}$
- **Now you can add:** $^{5}/_{30} + {}^{16}/_{30} = {}^{21}/_{30} = {}^{7}/_{10}$ (simplified)

Answer: LCD = 30

Why This Works: Even with larger numbers, the process is the same: find what number both denominators divide into evenly.

Example 3: Three Fractions ($^{1}/_{4} + {}^{1}/_{3} + {}^{1}/_{6}$)

> **Problem**
>
> Find the common denominator for $^{1}/_{4}$, $^{1}/_{3}$, and $^{1}/_{6}$

Step-by-Step Solution:

Step 1: List multiples of all three denominators

- Multiples of 4: 4, 8, **12**, 16, 20...
- Multiples of 3: 3, 6, 9, **12**, 15...
- Multiples of 6: 6, **12**, 18, 24...
- **LCD = 12** (first number in all three lists)

Step 2: Convert $^{1}/_{4}$ to 12ths

- $^{1}/_{4} \times {}^{3}/_{3} = {}^{3}/_{12}$

Step 3: Convert $^{1}/_{3}$ to 12ths

- $^{1}/_{3} \times {}^{4}/_{4} = {}^{4}/_{12}$

Step 4: Convert $^{1}/_{6}$ to 12ths

- Ask: "What times 6 equals 12?" → 2
- $^{1}/_{6} \times {}^{2}/_{2} = {}^{2}/_{12}$

Step 5: All three fractions now have denominator 12

- $^{1}/_{4} = {}^{3}/_{12}$
- $^{1}/_{3} = {}^{4}/_{12}$
- $^{1}/_{6} = {}^{2}/_{12}$
- **Now you can add:** $^{3}/_{12} + {}^{4}/_{12} + {}^{2}/_{12} = {}^{9}/_{12} = {}^{3}/_{4}$ (simplified)

Answer: LCD = 12

Why This Works: When you have multiple fractions, find the LCD that works for ALL denominators. The LCD must be divisible by each denominator.

Practice Problems: Finding Common Denominators

Try These Yourself

1. Find LCD for $\frac{1}{2}$ and $\frac{1}{3}$
2. Find LCD for $\frac{2}{3}$ and $\frac{3}{4}$
3. Find LCD for $\frac{1}{5}$, $\frac{2}{5}$, and $\frac{3}{10}$
4. Find LCD for $\frac{3}{8}$ and $\frac{5}{12}$
5. Find LCD for $\frac{1}{6}$, $\frac{1}{4}$, and $\frac{1}{8}$

Answer Key: Finding Common Denominators

Solutions

1. Find LCD for $\frac{1}{2}$ and $\frac{1}{3}$

Solution:

Multiples of 2: 2, 4, 6, 8...

Multiples of 3: 3, 6, 9...

LCD = 6

$\frac{1}{2} = \frac{3}{6}$, $\frac{1}{3} = \frac{2}{6}$

Answer: LCD = 6

2. Find LCD for $\frac{2}{3}$ and $\frac{3}{4}$

Solution:

Multiples of 3: 3, 6, 9, 12...

Multiples of 4: 4, 8, 12...

LCD = 12

$\frac{2}{3} = \frac{8}{12}$, $\frac{3}{4} = \frac{9}{12}$

Answer: LCD = 12

3. Find LCD for $\frac{1}{5}$, $\frac{2}{5}$, and $\frac{3}{10}$

Solution:

Multiples of 5: 5, 10, 15...

Multiples of 10: 10, 20...

LCD = 10

$\frac{1}{5} = \frac{2}{10}$, $\frac{2}{5} = \frac{4}{10}$, $\frac{3}{10} = \frac{3}{10}$

Answer: LCD = 10

4. Find LCD for $\frac{3}{8}$ and $\frac{5}{12}$

Solution:

Multiples of 8: 8, 16, 24...

Multiples of 12: 12, 24...

LCD = 24

$\frac{3}{8} = \frac{9}{24}$, $\frac{5}{12} = \frac{10}{24}$

Answer: LCD = 24

5. Find LCD for $\frac{1}{6}$, $\frac{1}{4}$, and $\frac{1}{8}$

Solution:

Multiples of 6: 6, 12, 18, 24...

Multiples of 4: 4, 8, 12, 16, 20, 24...

Multiples of 8: 8, 16, 24...

LCD = 24

$\frac{1}{6} = \frac{4}{24}$, $\frac{1}{4} = \frac{6}{24}$, $\frac{1}{8} = \frac{3}{24}$

Answer: LCD = 24

Method 2: Prime Factorization (For Complex Fractions)

Example: Find LCD for $\frac{1}{6}$ and $\frac{8}{15}$

Step 1: Factor denominators into primes

- $6 = 2 \times 3$
- $15 = 3 \times 5$

Step 2: Take highest power of each prime

- Prime factors: 2, 3, 5
- Highest powers: 2^1, 3^1, 5^1

Step 3: Multiply: 2 × 3 × 5 = 30

- **LCD = 30**

Step 4: Convert fractions

- $\frac{1}{6}$ = ?/30 → Multiply by 5 → $\frac{5}{30}$
- $\frac{8}{15}$ = ?/30 → Multiply by 2 → $\frac{16}{30}$

Adding Fractions: Why Common Denominators Matter

🔵 The Rule

Add numerators, keep denominator the same

But ONLY after finding common denominator!

Why Common Denominators?

```
You can't add: ½ + ⅓ directly
Because: "half" and "third" are different sized pieces!

Visual:
[████] [    ]  +  [██] [██] [  ] = ???
 ½       ½         ⅓   ⅓   ⅓

Convert to same size pieces:
[████] [████] [    ]  +  [████] [████] [████] = [████] [████] [████]
[████] [    ]
 3/6     3/6     3/6      2/6      2/6      2/6        5/6
```

Example 1: Basic Addition (½ + ¼)

Problem

Add ½ + ¼

Step-by-Step Solution:

Step 1: Check if denominators are the same

- ½ has denominator 2
- ¼ has denominator 4
- They're different, so we need a common denominator

Step 2: Find LCD

- Multiples of 2: 2, **4**, 6...
- Multiples of 4: **4**, 8...
- LCD = 4

Step 3: Convert ½ to have denominator 4

- $\frac{1}{2} \times \frac{2}{2} = \frac{2}{4}$

Step 4: Convert ¼ (already has denominator 4)

- $\frac{1}{4} = \frac{1}{4}$ (no change needed)

Step 5: Add numerators, keep denominator

- $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$

Answer: ¾

Why This Works: Once fractions have the same denominator, you're adding pieces of the same size. Just add how many pieces you have total.

Example 2: Different Denominators (⅔ + ¾)

> **Problem**
>
> Add ⅔ + ¾

Step-by-Step Solution:

Step 1: Find LCD

- Multiples of 3: 3, 6, 9, **12**...
- Multiples of 4: 4, 8, **12**...
- LCD = 12

Step 2: Convert ⅔ to 12ths

- Ask: "What times 3 equals 12?" → 4
- $\frac{2}{3} \times \frac{4}{4} = \frac{8}{12}$

Step 3: Convert ¾ to 12ths

- Ask: "What times 4 equals 12?" → 3
- $\frac{3}{4} \times \frac{3}{3} = \frac{9}{12}$

Step 4: Add numerators

- 8 + 9 = 17

Step 5: Keep denominator

- $^{17}/_{12}$

Step 6: Simplify (if needed)

- $^{17}/_{12}$ = 1 $^{5}/_{12}$ (mixed number)

Answer: 1 $^{5}/_{12}$ or $^{17}/_{12}$

Why This Works: Both fractions are now in "twelfths" - same size pieces. Adding 8 twelfths + 9 twelfths = 17 twelfths.

Example 3: Mixed Numbers

> **Problem**
>
> Add 2½ + 1⅓

Step-by-Step Solution:

Step 1: Convert mixed numbers to improper fractions

- 2½ = $^{5}/_{2}$ (2 × 2 + 1 = 5)
- 1⅓ = $^{4}/_{3}$ (1 × 3 + 1 = 4)

Step 2: Find LCD for $^{5}/_{2}$ and $^{4}/_{3}$

- Multiples of 2: 2, 4, **6**...
- Multiples of 3: 3, **6**...
- LCD = 6

Step 3: Convert $^{5}/_{2}$ to 6ths

- $^{5}/_{2} \times {}^{3}/_{3} = {}^{15}/_{6}$

Step 4: Convert $^{4}/_{3}$ to 6ths

- $^{4}/_{3} \times {}^{2}/_{2} = {}^{8}/_{6}$

Step 5: Add

- $^{15}/_{6} + {}^{8}/_{6} = {}^{23}/_{6}$

Step 6: Simplify to mixed number

- 23 ÷ 6 = 3 remainder 5

- $^{23}/_{6} = 3\ ^{5}/_{6}$

Answer: $3\ ^{5}/_{6}$

Why This Works: Convert mixed numbers to improper fractions first, then add normally. Convert back to mixed number at the end.

Practice Problems: Adding Fractions

Try These Yourself

1. $\frac{1}{4} + \frac{1}{3} = ?$
2. $\frac{2}{3} + \frac{5}{6} = ?$
3. $1\frac{1}{2} + 2\frac{3}{4} = ?$
4. $\frac{3}{8} + \frac{5}{8} = ?$
5. $\frac{1}{5} + \frac{2}{5} + \frac{3}{10} = ?$

Answer Key: Adding Fractions

Solutions

1. $\frac{1}{4} + \frac{1}{3} = ?$

Solution:

LCD = 12

$\frac{1}{4} = \frac{3}{12}$, $\frac{1}{3} = \frac{4}{12}$

$\frac{3}{12} + \frac{4}{12} = \frac{7}{12}$

Answer: $\frac{7}{12}$

2. $\frac{2}{3} + \frac{5}{6} = ?$

Solution:

LCD = 6

$\frac{2}{3} = \frac{4}{6}$, $\frac{5}{6} = \frac{5}{6}$

$\frac{4}{6} + \frac{5}{6} = \frac{9}{6} = 1\frac{1}{2}$

Answer: $1\frac{1}{2}$ or $\frac{3}{2}$

3. $1\frac{1}{2} + 2\frac{3}{4} = ?$

Solution:

Convert: $1\frac{1}{2} = \frac{3}{2}$, $2\frac{3}{4} = \frac{11}{4}$

LCD = 4

$\frac{3}{2} = \frac{6}{4}$, $\frac{11}{4} = \frac{11}{4}$

$\frac{6}{4} + \frac{11}{4} = \frac{17}{4} = 4\frac{1}{4}$

Answer: $4\frac{1}{4}$

4. $\frac{3}{8} + \frac{5}{8} = ?$

Solution:

Same denominator, just add numerators

3 + 5 = 8

$\frac{3}{8} + \frac{5}{8} = \frac{8}{8} = 1$

Answer: 1

5. $\frac{1}{5} + \frac{2}{5} + \frac{3}{10} = ?$

Solution:

LCD = 10

$\frac{1}{5} = \frac{2}{10}$, $\frac{2}{5} = \frac{4}{10}$, $\frac{3}{10} = \frac{3}{10}$

$\frac{2}{10} + \frac{4}{10} + \frac{3}{10} = \frac{9}{10}$

Answer: $\frac{9}{10}$

Formula:

```
a/b + c/d = (ad + bc) / bd
(But finding LCD is usually easier!)
```

Subtracting Fractions

🟠 Same Process as Addition

Find common denominator first, then subtract numerators

Example 1: Basic Subtraction ($\frac{2}{3} - \frac{1}{4}$)

Problem

Subtract $\frac{2}{3} - \frac{1}{4}$

Step-by-Step Solution:

Step 1: Find LCD

- Multiples of 3: 3, 6, 9, **12**...
- Multiples of 4: 4, 8, **12**...
- LCD = 12

Step 2: Convert $\frac{2}{3}$ to 12ths

- $\frac{2}{3} \times \frac{4}{4} = \frac{8}{12}$

Step 3: Convert $\frac{1}{4}$ to 12ths

- $\frac{1}{4} \times \frac{3}{3} = \frac{3}{12}$

Step 4: Subtract numerators

- 8 - 3 = 5

Step 5: Keep denominator

- $\frac{5}{12}$

Answer: $\frac{5}{12}$

Why This Works: Same as addition - once denominators match, subtract the numerators to find the difference.

Example 2: With Borrowing ($1 - \frac{3}{5}$)

Problem

Subtract $1 - \frac{3}{5}$

Step-by-Step Solution:

Step 1: Convert whole number to fraction

- $1 = \frac{1}{1}$

Step 2: Find LCD

- Multiples of 1: 1, 2, 3, 4, **5**...
- Multiples of 5: **5**, 10...

- LCD = 5

Step 3: Convert $\frac{1}{1}$ to 5ths

- $\frac{1}{1} \times \frac{5}{5} = \frac{5}{5}$

Step 4: Convert $\frac{3}{5}$ (already in 5ths)

- $\frac{3}{5} = \frac{3}{5}$

Step 5: Subtract

- $\frac{5}{5} - \frac{3}{5} = \frac{2}{5}$

Answer: $\frac{2}{5}$

Why This Works: Think of 1 as $\frac{5}{5}$ (5 out of 5 parts). Subtracting 3 parts leaves 2 parts.

Example 3: Mixed Numbers

> **Problem**
>
> Subtract $3\frac{1}{3} - 1\frac{3}{4}$

Step-by-Step Solution:

Step 1: Convert to improper fractions

- $3\frac{1}{3} = \frac{10}{3}$ (3 × 3 + 1 = 10)
- $1\frac{3}{4} = \frac{7}{4}$ (1 × 4 + 3 = 7)

Step 2: Find LCD

- Multiples of 3: 3, 6, 9, **12**...
- Multiples of 4: 4, 8, **12**...
- LCD = 12

Step 3: Convert $\frac{10}{3}$ to 12ths

- $\frac{10}{3} \times \frac{4}{4} = \frac{40}{12}$

Step 4: Convert $\frac{7}{4}$ to 12ths

- $\frac{7}{4} \times \frac{3}{3} = \frac{21}{12}$

Step 5: Subtract

- $\frac{40}{12} - \frac{21}{12} = \frac{19}{12}$

Step 6: Convert to mixed number

- 19 ÷ 12 = 1 remainder 7

- $^{19}/_{12} = 1\,^{7}/_{12}$

Answer: $1\,^{7}/_{12}$

Why This Works: Convert mixed numbers first, then subtract normally. Convert back to mixed number at the end.

Practice Problems: Subtracting Fractions

Try These Yourself

1. $^{5}/_{6} - ^{1}/_{3} = ?$
2. $^{3}/_{4} - ^{1}/_{2} = ?$
3. $2 - ^{3}/_{5} = ?$
4. $2^{2}/_{3} - 1^{1}/_{4} = ?$
5. $^{7}/_{8} - ^{3}/_{8} = ?$

Answer Key: Subtracting Fractions

Solutions

1. $^{5}/_{6} - ^{1}/_{3} = ?$

Solution:

LCD = 6

$^{5}/_{6} = ^{5}/_{6}$, $^{1}/_{3} = ^{2}/_{6}$

$^{5}/_{6} - ^{2}/_{6} = ^{3}/_{6} = ^{1}/_{2}$

Answer: $^{1}/_{2}$

2. $^{3}/_{4} - ^{1}/_{2} = ?$

Solution:

LCD = 4

$^{3}/_{4} = ^{3}/_{4}$, $^{1}/_{2} = ^{2}/_{4}$

$^{3}/_{4} - ^{2}/_{4} = ^{1}/_{4}$

Answer: $^{1}/_{4}$

3. 2 - $\frac{3}{5}$ = ?

Solution:

2 = $\frac{10}{5}$

$\frac{10}{5}$ - $\frac{3}{5}$ = $\frac{7}{5}$ = $1\frac{2}{5}$

Answer: $1\frac{2}{5}$

4. $2\frac{2}{3}$ - $1\frac{1}{4}$ = ?

Solution:

Convert: $2\frac{2}{3}$ = $\frac{8}{3}$, $1\frac{1}{4}$ = $\frac{5}{4}$

LCD = 12

$\frac{8}{3}$ = $\frac{32}{12}$, $\frac{5}{4}$ = $\frac{15}{12}$

$\frac{32}{12}$ - $\frac{15}{12}$ = $\frac{17}{12}$ = $1\frac{5}{12}$

Answer: $1\frac{5}{12}$

5. $\frac{7}{8}$ - $\frac{3}{8}$ = ?

Solution:

Same denominator

7 - 3 = 4

$\frac{7}{8}$ - $\frac{3}{8}$ = $\frac{4}{8}$ = $\frac{1}{2}$

Answer: $\frac{1}{2}$

Multiplying Fractions: Why Multiply Top and Bottom

🟢 The Rule

Multiply numerators, multiply denominators

No common denominator needed!

Why It Works:

```
½ × ⅓ = ?

Think: "Half of a third"
Visual: Take ⅓, then take half of it = ⅙

Formula: (1 × 1) / (2 × 3) = 1/6 ✓
```

Example 1: Simple Multiplication (½ × ⅓)

> **Problem**
>
> Multiply ½ × ⅓

Step-by-Step Solution:

Step 1: Multiply numerators

- 1 × 1 = 1

Step 2: Multiply denominators

- 2 × 3 = 6

Step 3: Write the answer

- $\frac{1}{6}$

Step 4: Check if it can be simplified

- $\frac{1}{6}$ is already in simplest form

Answer: ⅙

Why This Works: "Half of a third" means take ⅓ and divide it in half. This gives you ⅙ of the whole.

Example 2: Mixed Numbers (2½ × 1⅓)

> **Problem**
>
> Multiply 2½ × 1⅓

Step-by-Step Solution:

Step 1: Convert mixed numbers to improper fractions

- 2½ = $\frac{5}{2}$ (2 × 2 + 1 = 5)
- 1⅓ = $\frac{4}{3}$ (1 × 3 + 1 = 4)

Step 2: Multiply numerators

- 5 × 4 = 20

Step 3: Multiply denominators

- $2 \times 3 = 6$

Step 4: Write the answer

- $^{20}/_{6}$

Step 5: Simplify

- $20 \div 2 = 10$, $6 \div 2 = 3$
- $^{20}/_{6} = {}^{10}/_{3}$

Step 6: Convert to mixed number

- $10 \div 3 = 3$ remainder 1
- $^{10}/_{3} = 3\frac{1}{3}$

Answer: $3\frac{1}{3}$

Why This Works: Always convert mixed numbers to improper fractions first, then multiply normally.

Example 3: Word Problem Application

> **Problem**
>
> A recipe calls for $\frac{2}{3}$ cup of flour. If you want to make $\frac{3}{4}$ of the recipe, how much flour do you need?

Step-by-Step Solution:

Step 1: Identify what you're finding

- Original amount: $\frac{2}{3}$ cup
- Fraction of recipe: $\frac{3}{4}$
- Need: $\frac{2}{3} \times \frac{3}{4}$

Step 2: Multiply numerators

- $2 \times 3 = 6$

Step 3: Multiply denominators

- $3 \times 4 = 12$

Step 4: Write answer

- $^{6}/_{12}$

Step 5: Simplify

- $6 \div 6 = 1$, $12 \div 6 = 2$
- $^{6}/_{12} = \frac{1}{2}$

Answer: ½ cup

Why This Works: "¾ of ⅔" means multiply. You're taking three-fourths of two-thirds.

Practice Problems: Multiplying Fractions

Try These Yourself

1. ⅔ × ¾ = ?
2. ½ × ⅘ = ?
3. 1½ × 2⅓ = ?
4. ⅜ × ⅘ = ?
5. If you have ⅔ of a pizza and eat ½ of that, how much did you eat?

Answer Key: Multiplying Fractions

Solutions

1. ⅔ × ¾ = ?

Solution:

Numerators: 2 × 3 = 6

Denominators: 3 × 4 = 12

⁶⁄₁₂ = ½

Answer: ½

2. ½ × ⅘ = ?

Solution:

Numerators: 1 × 4 = 4

Denominators: 2 × 5 = 10

⁴⁄₁₀ = ⅖

Answer: ⅖

3. 1½ × 2⅓ = ?

Solution:

Convert: $1\frac{1}{2} = \frac{3}{2}$, $2\frac{1}{3} = \frac{7}{3}$

$\frac{3}{2} \times \frac{7}{3} = \frac{21}{6} = \frac{7}{2} = 3\frac{1}{2}$

Answer: $3\frac{1}{2}$

4. $\frac{3}{8} \times \frac{4}{5} = ?$

Solution:

Numerators: 3 × 4 = 12

Denominators: 8 × 5 = 40

$\frac{12}{40} = \frac{3}{10}$

Answer: $\frac{3}{10}$

5. If you have $\frac{2}{3}$ of a pizza and eat $\frac{1}{2}$ of that, how much did you eat?

Solution:

$\frac{2}{3} \times \frac{1}{2} = \frac{2}{6} = \frac{1}{3}$

Answer: $\frac{1}{3}$ of the pizza

Memory Tip: "Top times top, bottom times bottom"

Dividing Fractions: Why Flip and Multiply

🔴 The Rule

Flip the second fraction, then multiply

"Keep, Change, Flip"

Why Flip?

```
½ ÷ ⅓ = ?

Think: "How many thirds fit into a half?"
Actually easier: "How many times does ⅓ go into ½?"

Better way: ½ ÷ ⅓ = ½ × ³/₁ = ³/₂ = 1½
(Flip ⅓ to ³/₁, then multiply)
```

Example 1: Basic Division ($\frac{1}{2} \div \frac{1}{3}$)

Problem

> Divide $\frac{1}{2} \div \frac{1}{3}$

Step-by-Step Solution:

Step 1: Keep the first fraction

- $\frac{1}{2}$ stays as $\frac{1}{2}$

Step 2: Change ÷ to ×

- $\frac{1}{2} \div \frac{1}{3}$ becomes $\frac{1}{2} \times ?$

Step 3: Flip the second fraction (find reciprocal)

- $\frac{1}{3}$ flipped = $\frac{3}{1}$ (swap numerator and denominator)

Step 4: Multiply

- $\frac{1}{2} \times \frac{3}{1} = \frac{3}{2}$

Step 5: Simplify or convert to mixed number

- $\frac{3}{2} = 1\frac{1}{2}$

Answer: $1\frac{1}{2}$

Why This Works: Dividing by $\frac{1}{3}$ is the same as multiplying by 3. "How many thirds fit into a half?" Answer: 1.5 thirds.

Example 2: Complex Division ($2\frac{2}{3} \div 1\frac{1}{4}$)

> **Problem**
>
> Divide $2\frac{2}{3} \div 1\frac{1}{4}$

Step-by-Step Solution:

Step 1: Convert mixed numbers to improper fractions

- $2\frac{2}{3} = \frac{8}{3}$ ($2 \times 3 + 2 = 8$)
- $1\frac{1}{4} = \frac{5}{4}$ ($1 \times 4 + 1 = 5$)

Step 2: Keep, Change, Flip

- Keep: $\frac{8}{3}$
- Change: ÷ becomes ×
- Flip: $\frac{5}{4}$ becomes $\frac{4}{5}$

Step 3: Multiply

- $^{8}/_{3} \times {}^{4}/_{5} = {}^{32}/_{15}$

Step 4: Simplify

- 32 and 15 share no common factors

Step 5: Convert to mixed number

- 32 ÷ 15 = 2 remainder 2
- $^{32}/_{15} = 2\ ^{2}/_{15}$

Answer: $2\ ^{2}/_{15}$

Why This Works: Always convert mixed numbers first, then use "Keep, Change, Flip" to divide.

Example 3: Word Problem Application

> **Problem**
>
> You have 3½ yards of fabric. Each project needs ⅞ yard. How many projects can you make?

Step-by-Step Solution:

Step 1: Set up the division

- Total fabric: 3½ yards
- Fabric per project: ⅞ yard
- Number of projects = 3½ ÷ ⅞

Step 2: Convert mixed number

- 3½ = $^{7}/_{2}$

Step 3: Keep, Change, Flip

- Keep: $^{7}/_{2}$
- Change: ÷ becomes ×
- Flip: ⅞ becomes $^{8}/_{7}$

Step 4: Multiply

- $^{7}/_{2} \times {}^{8}/_{7} = {}^{56}/_{14}$

Step 5: Simplify

- 56 ÷ 14 = 4
- $^{56}/_{14} = 4$

Answer: 4 projects

Why This Works: Dividing total by amount per item tells you how many items you can make.

Practice Problems: Dividing Fractions

Try These Yourself

1. $\frac{2}{3} \div \frac{1}{4} = ?$
2. $\frac{3}{4} \div \frac{1}{2} = ?$
3. $2 \div \frac{1}{3} = ?$
4. $1\frac{1}{3} \div \frac{2}{3} = ?$
5. If you have $2\frac{1}{2}$ cups of juice and each serving is $\frac{5}{8}$ cup, how many servings?

Answer Key: Dividing Fractions

Solutions

1. $\frac{2}{3} \div \frac{1}{4} = ?$

Solution:

Keep: $\frac{2}{3}$, Change: $\div$ to $\times$, Flip: $\frac{1}{4}$ to $\frac{4}{1}$

$\frac{2}{3} \times \frac{4}{1} = \frac{8}{3} = 2\frac{2}{3}$

Answer: $2\frac{2}{3}$

2. $\frac{3}{4} \div \frac{1}{2} = ?$

Solution:

Keep: $\frac{3}{4}$, Change: $\div$ to $\times$, Flip: $\frac{1}{2}$ to $\frac{2}{1}$

$\frac{3}{4} \times \frac{2}{1} = \frac{6}{4} = 1\frac{1}{2}$

Answer: $1\frac{1}{2}$

3. $2 \div \frac{1}{3} = ?$

Solution:

$2 = \frac{2}{1}$

$\frac{2}{1} \div \frac{1}{3} = \frac{2}{1} \times \frac{3}{1} = \frac{6}{1} = 6$

Answer: 6

4. $1\frac{1}{3} \div \frac{2}{3} = ?$

Solution:

Convert: $1\frac{1}{3} = \frac{4}{3}$

$\frac{4}{3} \div \frac{2}{3} = \frac{4}{3} \times \frac{3}{2} = \frac{12}{6} = 2$

Answer: 2

5. If you have 2½ cups of juice and each serving is ⅝ cup, how many servings?

Solution:

$2\frac{1}{2} = \frac{5}{2}$

$\frac{5}{2} \div \frac{5}{8} = \frac{5}{2} \times \frac{8}{5} = \frac{40}{10} = 4$

Answer: 4 servings

Memory Tip: "Keep, Change, Flip" or "Multiply by the reciprocal"

Converting Fractions to Decimals

> 🔵 Method: Divide Numerator by Denominator
>
> Fraction = Division problem!

Examples:

Simple Fractions:

- $\frac{1}{2} = 1 \div 2 = 0.5$
- $\frac{1}{4} = 1 \div 4 = 0.25$
- $\frac{3}{4} = 3 \div 4 = 0.75$

Common Fractions to Memorize:

- $\frac{1}{2} = 0.5$
- $\frac{1}{3} = 0.333...$
- $\frac{2}{3} = 0.666...$
- $\frac{1}{4} = 0.25$
- $\frac{3}{4} = 0.75$
- $\frac{1}{5} = 0.2$
- $\frac{2}{5} = 0.4$
- $\frac{3}{5} = 0.6$
- $\frac{4}{5} = 0.8$

Example 1: Simple Conversion (½ = ?)

> **Problem**

> Convert ½ to a decimal

Step-by-Step Solution:

Step 1: Set up division

- ½ means 1 ÷ 2
- Write as: 2) 1

Step 2: Add decimal point and zeros

- 2) 1.0

Step 3: Divide

- How many times does 2 go into 1? 0 times
- Write 0, then decimal point
- 2) 1.0
- 0.

Step 4: Bring down the 0

- 2) 1.0
- 0.
- 10

Step 5: Divide again

- How many times does 2 go into 10? 5 times
- Write 5
- 2) 1.0
- 0.5
- 10
- -10
- ```
  0
  ```

Answer: 0.5

Why This Works: A fraction is division. ½ = 1 ÷ 2 = 0.5

Example 2: Complex Conversion (⅜ = ?) with Long Division

> **Problem**
  ```

> Convert $\frac{3}{8}$ to a decimal

Step-by-Step Solution:

Step 1: Set up division

- $\frac{3}{8} = 3 \div 8$
- 8) 3.000

Step 2: First digit (ones place)

- 8 doesn't go into 3, so write 0
- Add decimal point
- 8) 3.000
- 0.

Step 3: Bring down first 0 (tenths)

- 8) 3.000
- 0.
- 30

Step 4: Divide 30 by 8

- $8 \times 3 = 24$, remainder 6
- Write 3 in tenths place
- 8) 3.000
- 0.3
- 30
- -24
- 6

Step 5: Bring down next 0 (hundredths)

- 8) 3.000
- 0.3
- 30
- -24
- 60

Step 6: Divide 60 by 8

- 8 × 7 = 56, remainder 4
- Write 7 in hundredths place
- 8) 3.000
- 0.37
- 30
- -24
- ```
  60
  ```
- -56
- ```
  4
  ```

Step 7: Bring down last 0 (thousandths)

- 8) 3.000
- 0.37
- 30
- -24
- ```
  60
  ```
- -56
- ```
  40
  ```

Step 8: Divide 40 by 8

- 8 × 5 = 40, no remainder
- Write 5 in thousandths place
- 8) 3.000
- 0.375
- 30
- -24
- ```
  60
  ```
- -56
- ```
  40
  ```

- ```
  -40
  ```
- ```
  0
  ```

Answer: 0.375

Why This Works: Long division shows each decimal place step-by-step. Keep bringing down zeros until you get a remainder of 0 or see a repeating pattern.

Example 3: Repeating Decimals (⅓ = ?)

> **Problem**
>
> Convert ⅓ to a decimal

Step-by-Step Solution:

Step 1: Set up division

- ⅓ = 1 ÷ 3
- 3) 1.000...

Step 2: Divide

- 3 doesn't go into 1, so 0.3...
- 3) 1.0
- 0.3
- 10
- -9
- ```
  1
  ```

Step 3: Notice the pattern

- Remainder is 1 again (same as start)
- This will repeat forever
- 3) 1.000...
- 0.333...
- 10
- -9
  ```

- ```
  10
  ```
- ```
  −9
  ```
- ```
  1 (repeats)
  ```

Answer: 0.333... or $0.\overline{3}$ (bar over 3 means repeating)

Why This Works: Some fractions create repeating decimals. When the remainder repeats, the decimal repeats.

Practice Problems: Converting Fractions to Decimals

Try These Yourself

1. ¼ = ?
2. ⅗ = ?
3. 5/16 = ?
4. ⅔ = ?
5. ⅞ = ?

Answer Key: Converting Fractions to Decimals

Solutions

1. ¼ = ?

Solution:

1 ÷ 4 = 0.25

Answer: 0.25

2. ⅗ = ?

Solution:
  ```

3 ÷ 5 = 0.6

Answer: 0.6

3. $^{5}/_{16}$ = ?

Solution:

5 ÷ 16 = 0.3125

Answer: 0.3125

4. $^{2}/_{3}$ = ?

Solution:

2 ÷ 3 = 0.666... = $0.\overline{6}$

Answer: $0.\overline{6}$ (repeating)

5. $^{7}/_{8}$ = ?

Solution:

7 ÷ 8 = 0.875

Answer: 0.875

Converting Decimals to Fractions

🟠 Method: Read the Decimal Place

Write as fraction, then simplify

Example 1: Terminating Decimal (0.75 = ?)

Problem

Convert 0.75 to a fraction

Step-by-Step Solution:

Step 1: Identify the decimal place

- 0.75 has 2 decimal places (hundredths place)

Step 2: Read it as a fraction

- "Seventy-five hundredths" = $^{75}/_{100}$

Step 3: Write as fraction

- $0.75 = {}^{75}/_{100}$

Step 4: Simplify by finding GCF

- Factors of 75: 1, 3, 5, 15, 25, 75
- Factors of 100: 1, 2, 4, 5, 10, 20, 25, 50, 100
- GCF = 25

Step 5: Divide numerator and denominator by GCF

- 75 ÷ 25 = 3
- 100 ÷ 25 = 4
- ${}^{75}/_{100} = {}^{3}/_{4}$

Answer: ¾

Why This Works: The decimal place tells you the denominator. Two decimal places = hundredths (100).

Example 2: Complex Decimal (0.375 = ?)

> **Problem**
>
> Convert 0.375 to a fraction

Step-by-Step Solution:

Step 1: Count decimal places

- 0.375 has 3 decimal places (thousandths place)

Step 2: Write as fraction

- $0.375 = {}^{375}/_{1000}$

Step 3: Simplify

- Find GCF of 375 and 1000
- $375 = 3 \times 5^3 = 3 \times 125$
- $1000 = 2^3 \times 5^3 = 8 \times 125$
- GCF = 125

Step 4: Divide by GCF

- 375 ÷ 125 = 3
- 1000 ÷ 125 = 8
- ${}^{375}/_{1000} = {}^{3}/_{8}$

Answer: ⅜

Why This Works: Three decimal places = thousandths (1000). Always simplify to lowest terms.

Example 3: Decimal with Simplification

Problem

Convert 0.6 to a fraction

Step-by-Step Solution:

Step 1: Count decimal places

- 0.6 has 1 decimal place (tenths place)

Step 2: Write as fraction

- 0.6 = $^{6}/_{10}$

Step 3: Simplify

- GCF of 6 and 10 = 2
- 6 ÷ 2 = 3
- 10 ÷ 2 = 5
- $^{6}/_{10}$ = $^{3}/_{5}$

Answer: $^{3}/_{5}$

Why This Works: One decimal place = tenths (10). Always simplify when possible.

Practice Problems: Converting Decimals to Fractions

Try These Yourself

1. 0.25 = ?
2. 0.4 = ?
3. 0.125 = ?
4. 0.875 = ?
5. 0.2 = ?

Answer Key: Converting Decimals to Fractions

Solutions

1. 0.25 = ?

Solution:

0.25 = $^{25}/_{100}$ = ¼

Answer: ¼

2. 0.4 = ?

Solution:

0.4 = $^{4}/_{10}$ = ⅖

Answer: ⅖

3. 0.125 = ?

Solution:

0.125 = $^{125}/_{1000}$ = ⅛

Answer: ⅛

4. 0.875 = ?

Solution:

0.875 = $^{875}/_{1000}$ = ⅞

Answer: ⅞

5. 0.2 = ?

Solution:

0.2 = $^{2}/_{10}$ = ⅕

Answer: ⅕

Quick Reference:

Decimal	Fraction	Simplified
0.5	$^{5}/_{10}$	½
0.25	$^{25}/_{100}$	¼
0.75	$^{75}/_{100}$	¾
0.2	$^{2}/_{10}$	⅕
0.4	$^{4}/_{10}$	⅖
0.6	$^{6}/_{10}$	⅗

Decimal	Fraction	Simplified
0.8	$^{8}/_{10}$	$^{4}/_{5}$

Simplifying Fractions

🟢 Find Greatest Common Factor (GCF)

Divide numerator and denominator by their GCF

Method 1: List Factors

Example: Simplify $^{12}/_{18}$

1. **Factors of 12:** 1, 2, 3, 4, 6, 12
2. **Factors of 18:** 1, 2, 3, 6, 9, 18
3. **GCF:** 6 (largest common factor)
4. **Divide:** 12 ÷ 6 = 2, 18 ÷ 6 = 3
5. **Answer:** $^{2}/_{3}$

Method 2: Prime Factorization

Example: Simplify $^{24}/_{36}$

1. **Prime factors of 24:** 2 × 2 × 2 × 3
2. **Prime factors of 36:** 2 × 2 × 3 × 3
3. **Common factors:** 2 × 2 × 3 = 12
4. **Divide:** 24 ÷ 12 = 2, 36 ÷ 12 = 3
5. **Answer:** $^{2}/_{3}$

3️⃣ Decimals & Percentages

Decimal Place Value

🟣 Understanding Decimal Places

Place	Value	Example
Ones	1	3
Tenths	0.1	0.3
Hundredths	0.01	0.03
Thousandths	0.001	0.003

Example: 3.456

- 3 = 3 ones
- 4 = 4 tenths (0.4)
- 5 = 5 hundredths (0.05)
- 6 = 6 thousandths (0.006)

What Percentages Really Mean

🔴 Percentage = "Parts Per 100"

Percent = "per cent" = "per 100"

50% means 50 out of 100, or 50/100, or 0.50

Visual Model:

```
50% = 50 out of 100 parts
[▓▓▓
```

- 150% → 1.50 (move 2 left)

Memory Tip:

- **D**ecimal to **P**ercent: **D**ecimal point moves **R**ight
- **P**ercent to **D**ecimal: **P**oint moves **L**eft

Why "Of" Means Multiply

🟠 The Key Concept

When you see "X% of Y", it means "X% times Y"

"Of" = multiplication!

Why?

```
"50% of 200" means "50 out of every 100 of 200"

Think: Half of 200 = 100
Formula: 50% × 200 = 0.50 × 200 = 100 ✓
```

Examples:

- "12% of $375" = 0.12 × $375 = $45
- "25% of 80" = 0.25 × 80 = 20
- "10% of $8.50" = 0.10 × $8.50 = $0.85

Memory Tip: "Of" = "times" (multiplication)

Percentage Calculations

🟢 Three Types of Problems

Type 1: Find the Part

Given: Percentage and Whole
Find: Part
Formula: Part = Percentage × Whole

Example 1: Finding Part (12% of $375)

Problem

> A waitress earns an average tip of 12% of the cost of food. If she serves $375 worth of food, how much in tips does she earn?

Step-by-Step Solution:

Step 1: Identify what you know

- Percentage: 12%
- Whole: $375
- Find: Part (tip amount)

Step 2: Convert percentage to decimal

- 12% = 12 ÷ 100 = 0.12

Step 3: Apply formula

- Part = Percentage × Whole
- Part = 0.12 × $375

Step 4: Multiply

- 0.12 × 375 = 45

Answer: $45.00

Why This Works: "12% of" means multiply. Convert percent to decimal first, then multiply.

Example 2: Finding Part (25% of 80)

> **Problem**
>
> What is 25% of 80?

Step-by-Step Solution:

Step 1: Convert percentage

- 25% = 0.25

Step 2: Multiply

- 0.25 × 80 = 20

Answer: 20

Why This Works: 25% is the same as ¼, and ¼ of 80 = 20. The formula works the same way.

Type 2: Find the Percentage

Given: Part and Whole
Find: Percentage
Formula: Percentage = (Part ÷ Whole) × 100

Example 1: Finding Percentage (40 out of 60)

> **Problem**
>
> If you got 40 questions correct out of 60 total, what percentage did you score?

Step-by-Step Solution:

Step 1: Identify what you know

- Part: 40 correct
- Whole: 60 total
- Find: Percentage

Step 2: Apply formula

- Percentage = (Part ÷ Whole) × 100
- Percentage = (40 ÷ 60) × 100

Step 3: Divide first

- 40 ÷ 60 = 0.6667

Step 4: Multiply by 100

- 0.6667 × 100 = 66.67%

Answer: 66.67% (or 66⅔%)

Why This Works: Divide part by whole to get the decimal, then multiply by 100 to convert to percent.

Example 2: Finding Percentage (15 out of 20)

> **Problem**
>
> 15 out of 20 students passed the test. What percentage passed?

Step-by-Step Solution:

Step 1: Apply formula

- Percentage = (15 ÷ 20) × 100

Step 2: Divide

- 15 ÷ 20 = 0.75

Step 3: Convert to percent

- 0.75 × 100 = 75%

Answer: 75%

Why This Works: 15/20 = ¾ = 0.75 = 75%. The formula gives the same result.

Type 3: Find the Whole

Given: Part and Percentage
Find: Whole
Formula: Whole = Part ÷ Percentage

Example 1: Finding Whole (600,000 is 120% of ?)

> **Problem**
>
> 600,000 is 120% of what number?

Step-by-Step Solution:

Step 1: Identify what you know

- Part: 600,000
- Percentage: 120%
- Find: Whole (original number)

Step 2: Convert percentage to decimal

- 120% = 1.20

Step 3: Apply formula

- Whole = Part ÷ Percentage
- Whole = 600,000 ÷ 1.20

Step 4: Divide

- 600,000 ÷ 1.20 = 500,000

Answer: 500,000

Why This Works: If 600,000 is 120% of the original, the original must be smaller. Divide by the percentage (as decimal) to find it.

Example 2: Finding Original Price

Problem

After a 20% discount, a shirt costs $40. What was the original price?

Step-by-Step Solution:

Step 1: Understand the problem

- Final price: $40 (this is 80% of original, since 100% - 20% = 80%)
- Find: Original price (100%)

Step 2: Set up formula

- $40 is 80% of original
- Whole = Part ÷ Percentage
- Original = $40 ÷ 0.80

Step 3: Divide

- $40 ÷ 0.80 = $50

Answer: $50

Why This Works: The discounted price represents 80% of the original. Divide by 0.80 to find 100%.

Practice Problems: Percentage Calculations

Try These Yourself

1. What is 15% of 200?
2. What is 8% of $50?
3. 18 out of 24 = what percent?
4. 30 out of 50 = what percent?
5. If 45 is 90% of a number, what is the number?
6. If $72 is 80% of the original price, what was the original price?

Answer Key: Percentage Calculations

Solutions

1. What is 15% of 200?

Solution:

Part = 0.15 × 200 = 30

Answer: 30

2. What is 8% of $50?

Solution:

Part = 0.08 × $50 = $4

Answer: $4

3. 18 out of 24 = what percent?

Solution:

Percentage = (18 ÷ 24) × 100 = 0.75 × 100 = 75%

Answer: 75%

4. 30 out of 50 = what percent?

Solution:

Percentage = (30 ÷ 50) × 100 = 0.60 × 100 = 60%

Answer: 60%

5. If 45 is 90% of a number, what is the number?

Solution:

Whole = 45 ÷ 0.90 = 50

Answer: 50

6. If $72 is 80% of the original price, what was the original price?

Solution:

Whole = $72 ÷ 0.80 = $90

Answer: $90

Percentage Increase

🔴 Formula

New Amount = Original × (1 + Percentage)

Or: New Amount = Original + (Original × Percentage)

Example: 7% raise on $8.15/hour

Method 1:

1. Calculate raise: $8.15 × 0.07 = $0.57
2. Add to original: $8.15 + $0.57 = $8.72

Method 2:

1. Multiply by (1 + %): $8.15 × 1.07 = $8.72

Why Method 2 Works:

```
Original × (1 + 0.07) = Original × 1.07
This gives you 107% of original = original + 7% increase
```

Percentage Decrease

🔵 Formula

New Amount = Original × (1 - Percentage)

Example: 10% discount on $8.50

Method 1:

1. Calculate discount: $8.50 × 0.10 = $0.85
2. Subtract: $8.50 - $0.85 = $7.65

Method 2:

1. Multiply by (1 - %): $8.50 × 0.90 = $7.65

4️⃣ Unit Conversions: The Systematic Approach

Understanding Conversion Factors

🟠 Key Concept

A conversion factor is a fraction equal to 1 that changes units

Example: 12 inches / 1 foot = 1 (they're equal!)

Why Conversion Factors Work:

```
Multiplying by 1 doesn't change the value!
12 inches / 1 foot = 1 (true statement)
So: 3 feet × (12 inches / 1 foot) = 36 inches
```

When to Multiply vs Divide in Conversions

🟢 Decision Framework

Direction	Operation	Example
Larger → Smaller	Multiply	3 feet → inches: 3 × 12 = 36
Smaller → Larger	Divide	36 inches → feet: 36 ÷ 12 = 3

Memory Tip:

- **L**arger to **S**maller = **M**ultiply (L→S = M)
- **S**maller to **L**arger = **D**ivide (S→L = D)

Unit Cancellation Method (Dimensional Analysis)

🟣 The Foolproof Method

Set up conversion factors so units cancel, leaving desired unit

Example 1: Single-Step Conversion (3 feet to inches)

Problem

Convert 3 feet to inches

Step-by-Step Solution:

Step 1: Identify conversion factor

- 1 foot = 12 inches
- Conversion factor: 12 inches / 1 foot = 1

Step 2: Set up conversion

- 3 feet × (12 inches / 1 foot)

Step 3: Cancel units

- Feet cancel out, leaving inches

Step 4: Multiply

- $3 \times 12 = 36$

Answer: 36 inches

Why This Works: Larger unit (feet) to smaller unit (inches) = multiply. Each foot has 12 inches.

Example 2: Area Conversion (288 ft^2 to yd^2)

> **Problem**
>
> Convert 288 square feet to square yards

Step-by-Step Solution:

Step 1: Understand area conversion

- 1 yard = 3 feet
- 1 square yard = 3 ft × 3 ft = 9 square feet
- Conversion factor: 1 yd^2 / 9 ft^2 = 1

Step 2: Set up conversion (smaller to larger = divide)

- 288 ft^2 ÷ 9 ft^2/yd^2

Step 3: Divide

- 288 ÷ 9 = 32

Answer: 32 square yards

Why This Works: Smaller unit (ft^2) to larger unit (yd^2) = divide. Square yards are bigger than square feet.

Example 3: Multi-Step Conversion (3 feet, 8 inches to total inches)

> **Problem**
>
> Convert 3 feet, 8 inches to total inches

Step-by-Step Solution:

Step 1: Start with what you have

- 3 feet, 8 inches

Step 2: Convert feet to inches

- 3 feet × (12 inches / 1 foot) = 36 inches
- (feet cancel out!)

Step 3: Add remaining inches

- 36 inches + 8 inches = 44 inches

Answer: 44 inches

Why This Works: Convert the mixed units to a single unit, then add.

Example 4: Multi-Step Chain (3 tons to ounces)

> **Problem**
>
> Convert 3 tons to ounces

Step-by-Step Solution:

Step 1: Plan the conversion chain

- Tons → Pounds → Ounces

Step 2: Convert tons to pounds

- 1 ton = 2,000 pounds
- 3 tons × (2,000 lbs / 1 ton) = 6,000 lbs

Step 3: Convert pounds to ounces

- 1 pound = 16 ounces
- 6,000 lbs × (16 oz / 1 lb) = 96,000 oz

Step 4: Or do it all at once

- 3 tons × (2,000 lbs / 1 ton) × (16 oz / 1 lb)
- = 3 × 2,000 × 16 oz
- = 96,000 oz

Answer: 96,000 ounces

Why This Works: Chain conversions step-by-step, canceling units as you go. Each step multiplies by a conversion factor equal to 1.

Practice Problems: Unit Conversions

Try These Yourself

1. 5 feet = ? inches
2. 48 inches = ? feet
3. 180 square feet = ? square yards
4. 2 tons = ? pounds
5. 4 feet, 6 inches = ? total inches
6. 1.5 tons = ? ounces

Answer Key: Unit Conversions

Solutions

1. 5 feet = ? inches

Solution:

5 × 12 = 60

Answer: 60 inches

2. 48 inches = ? feet

Solution:

48 ÷ 12 = 4

Answer: 4 feet

3. 180 square feet = ? square yards

Solution:

180 ÷ 9 = 20

Answer: 20 square yards

4. 2 tons = ? pounds

Solution:

2 × 2,000 = 4,000

Answer: 4,000 pounds

5. 4 feet, 6 inches = ? total inches

Solution:

4 × 12 = 48, 48 + 6 = 54

Answer: 54 inches

6. 1.5 tons = ? ounces

Solution:

1.5 × 2,000 = 3,000 lbs

3,000 × 16 = 48,000 oz

Answer: 48,000 ounces

Common Conversion Factors

🔴 Essential Conversions to Memorize

Category	Conversion
Length	1 foot = 12 inches 1 yard = 3 feet 1 mile = 5,280 feet
Area	1 square yard = 9 square feet (1 yd^2 = 3 ft × 3 ft = 9 ft^2)
Weight	1 ton = 2,000 pounds 1 pound = 16 ounces
Time	1 hour = 60 minutes 1 minute = 60 seconds 1 day = 24 hours
Metric	1 meter = 100 cm 1 meter = 1,000 mm 1 km ≈ ⅝ mile
Other	1 dozen = 12 1 gross = 144

Multi-Step Conversions

🔵 Chain Conversions

Convert step-by-step, canceling units as you go

Example: Convert 3 tons to ounces

Step-by-Step:

1. **Tons to pounds:** 3 tons × (2,000 lbs / 1 ton) = 6,000 lbs
2. **Pounds to ounces:** 6,000 lbs × (16 oz / 1 lb) = 96,000 oz

Visual Chain:

```
3 tons → [× 2000] → 6,000 lbs → [× 16] → 96,000 oz
```

All at Once:

```
3 tons × (2000 lbs / 1 ton) × (16 oz / 1 lb)
= 3 × 2000 × 16 oz
= 96,000 oz
```

5️⃣ Ratios & Proportions

Understanding Ratios

🟠 What Ratios Represent

A ratio compares two quantities

Can be written as: 3:4 or ¾ or "3 to 4"

Examples:

- "3 apples for every 4 oranges" = 3:4
- "Speed of 60 miles per hour" = 60:1 or 60/1
- "2 parts water, 3 parts juice" = 2:3

Setting Up Proportions

🟢 Proportion = Two Equal Ratios

a/b = c/d (read: "a is to b as c is to d")

Example 1: Simple Proportion (3 apples cost $2, 9 apples cost ?)

> **Problem**
>
> If 3 apples cost $2, how much do 9 apples cost?

Step-by-Step Solution:

Step 1: Set up the proportion

- Known ratio: 3 apples / $2
- Unknown ratio: 9 apples / $x
- $\frac{3}{2} = \frac{9}{x}$ (keep units consistent!)

Step 2: Cross-multiply

- $3 \times x = 2 \times 9$
- $3x = 18$

Step 3: Solve for x

- $x = 18 \div 3$
- $x = 6$

Answer: $6

Why This Works: The ratio of apples to cost stays constant. If 3 apples cost $2, then 9 apples (3 times more) cost $6 (3 times more).

Example 2: Rate Problems (60 miles in 2 hours, ? miles in 5 hours)

> **Problem**
>
> If you travel 60 miles in 2 hours, how many miles will you travel in 5 hours at the same speed?

Step-by-Step Solution:

Step 1: Set up proportion

- Known: 60 miles / 2 hours
- Unknown: x miles / 5 hours
- $\frac{60}{2} = \frac{x}{5}$

Step 2: Cross-multiply

- $60 \times 5 = 2 \times x$

- 300 = 2x

Step 3: Solve

- x = 300 ÷ 2
- x = 150

Answer: 150 miles

Why This Works: Speed (rate) stays constant. Distance is proportional to time.

Example 3: Scale Problems

Problem

On a map, 1 inch represents 25 miles. If two cities are 3.5 inches apart on the map, how far apart are they in reality?

Step-by-Step Solution:

Step 1: Set up proportion

- Map scale: 1 inch / 25 miles
- Map distance: 3.5 inches / x miles
- $1/25 = 3.5/x$

Step 2: Cross-multiply

- 1 × x = 25 × 3.5
- x = 87.5

Answer: 87.5 miles

Why This Works: The scale ratio stays constant. Map distance is proportional to real distance.

Practice Problems: Ratios & Proportions

Try These Yourself

1. If 4 books cost $12, how much do 7 books cost?
2. If a car travels 150 miles in 3 hours, how far will it travel in 5 hours?
3. A recipe calls for 2 cups of flour for 12 cookies. How much flour for 30 cookies?
4. On a scale drawing, 2 cm = 10 meters. If something is 7 cm on the drawing, how long is it in reality?
5. If 5 workers can complete a job in 8 days, how long will it take 10 workers?

Answer Key: Ratios & Proportions

Solutions

1. If 4 books cost $12, how much do 7 books cost?

Solution:

$\frac{4}{12} = \frac{7}{x}$

$4x = 84$

$x = \$21$

Answer: $21

2. If a car travels 150 miles in 3 hours, how far will it travel in 5 hours?

Solution:

$\frac{150}{3} = \frac{x}{5}$

$150 \times 5 = 3x$

$750 = 3x$

$x = 250$ miles

Answer: 250 miles

3. A recipe calls for 2 cups of flour for 12 cookies. How much flour for 30 cookies?

Solution:

$\frac{2}{12} = \frac{x}{30}$

$2 \times 30 = 12x$

$60 = 12x$

$x = 5$ cups

Answer: 5 cups

4. On a scale drawing, 2 cm = 10 meters. If something is 7 cm on the drawing, how long is it in reality?

Solution:

$\frac{2}{10} = \frac{7}{x}$

2x = 70

x = 35 meters

Answer: 35 meters

5. If 5 workers can complete a job in 8 days, how long will it take 10 workers?

Solution:

This is inverse proportion (more workers = less time)

5 × 8 = 10 × x

40 = 10x

x = 4 days

Answer: 4 days

Cross-Multiplication: Why It Works

🟣 The Method

If a/b = c/d, then a × d = b × c

Why It Works:

```
If a/b = c/d, multiply both sides by b × d:

(a/b) × (b × d) = (c/d) × (b × d)
a × d = c × b

So: a × d = b × c ✓
```

Example: Solve $\frac{2}{3} = \frac{x}{12}$

1. **Cross-multiply:** 2 × 12 = 3 × x
2. **Simplify:** 24 = 3x
3. **Solve:** x = 8
4. **Check:** $\frac{2}{3} = \frac{8}{12}$ ✓ (both = 0.667)

Rate Problems as Ratios

🔴 Speed = Distance/Time Ratio

Rate = Distance ÷ Time

The Three Formulas:

```
Distance = Rate × Time
Rate = Distance ÷ Time
Time = Distance ÷ Rate
```

Memory Tip: "DIRT" - Distance Is Rate Times Time

Example: If you travel 60 miles in 2 hours, what's your speed?

Solution:

- Rate = Distance ÷ Time
- Rate = 60 miles ÷ 2 hours = 30 miles per hour

6️⃣ When to Multiply vs Divide: Decision Framework

Visual Decision Tree

🔵 Decision Framework

Question: What operation do I need?

1. Look for "per" + quantity:

→ "per mile", "per hour", "per person"

ANSWER: MULTIPLY

Example: $3.50 per mile × 12 miles = $42

2. Look for "how many" or "how much each":

→ "how many in each group", "how much per person"

ANSWER: DIVIDE

Example: 1,260 sq ft ÷ 35 people = 36 sq ft each

3. Look for "total" or "all together":

→ "total cost", "total distance", "sum"

ANSWER: MULTIPLY (if rate given) or ADD

Example: $450/month × 60 months = $27,000 total

4. Look for "average" or "per":

→ "average cost", "average speed"

ANSWER: DIVIDE

Example: $343.68 ÷ 3 months = $114.56 average

Common Word Problem Keywords

🟠 Keyword Guide

Operation	Keywords
Add	sum, total, combined, together, plus, and, more
Subtract	difference, less, minus, subtract, take away, remaining, fewer
Multiply	times, multiplied by, product, of (with %), per (with quantity), each
Divide	divided by, split, share equally, per (finding rate), average, quotient

---

"Per" Problems: Multiply or Divide?

🟢 The Rule

"X per Y" + quantity given → MULTIPLY

"X per Y" + total given → DIVIDE

Examples:

Multiply (finding total):

- "$3.50 per mile" × "12 miles" = $42 total
- "45 miles per hour" × "3.5 hours" = 157.5 miles total

Divide (finding rate):

- "157.5 miles" ÷ "3 hours" = 52.5 miles per hour
- "$343.68 total" ÷ "3 months" = $114.56 per month

7️⃣ Mental Math Strategies

Breaking Down Large Numbers

🟣 Strategy: Split and Combine

Break numbers into easier parts

Example: Multiply 12 × 15

Method 1: Break one number

```
12 × 15 = 12 × (10 + 5)
        = 12 × 10 + 12 × 5
        = 120 + 60
        = 180
```

Method 2: Use known facts

```
12 × 15 = (12 × 10) + (12 × 5)
        = 120 + 60
        = 180
```

Quick Percentage Calculations

🔴 Common Percentages

Percentage	Decimal	Fraction	Quick Method
10%	0.10	$\frac{1}{10}$	Move decimal 1 left
25%	0.25	$\frac{1}{4}$	Divide by 4
50%	0.50	$\frac{1}{2}$	Divide by 2
75%	0.75	$\frac{3}{4}$	Multiply by 3, divide by 4

Quick Calculations:

- 10% of 375 = 37.5 (move decimal)
- 25% of 80 = 20 (80 ÷ 4)
- 50% of 200 = 100 (200 ÷ 2)
- 12% of 375 = 10% + 2% = 37.5 + 7.5 = 45

Estimation Techniques

🔵 Rounding Strategy

> Round to nearest easy number, calculate, then adjust

Example: Estimate 48 × 7

Method:

1. Round: 48 ≈ 50
2. Calculate: 50 × 7 = 350
3. Adjust: 48 is 2 less than 50, so subtract 2 × 7 = 14
4. Answer: 350 - 14 = 336

Actual: 48 × 7 = 336 ✓

Fraction Shortcuts

🟠 Common Fraction Operations

Operation	Shortcut
½ of number	Divide by 2
¼ of number	Divide by 4
⅓ of number	Divide by 3
⅖ of number	Multiply by 2, divide by 5

Example: Find ⅖ of 40

Method 1: 40 × ⅖ = (40 × 2) ÷ 5 = 80 ÷ 5 = 16

Method 2: 40 ÷ 5 = 8, then 8 × 2 = 16

📚 Quick Reference: Essential Rules

🟢 Fundamental Rules

🔴 **Fractions**

- Add/Subtract: Need common denominator
- Multiply: Top × top, bottom × bottom

- Divide: Flip second, then multiply
- LCD: List multiples or use prime factors

🔵 Percentages

- "Of" = multiply
- % to decimal: move 2 left
- Decimal to %: move 2 right
- Increase: × (1 + %)
- Decrease: × (1 - %)

🟠 Conversions

- Larger → Smaller: Multiply
- Smaller → Larger: Divide
- Use unit cancellation
- Chain conversions step-by-step

🟣 Operations

- "Per" + quantity → Multiply
- "How many each" → Divide
- "Total" → Multiply or Add
- "Average" → Divide

🎓 Study Tips

📖 How to Master Fundamentals

1. **Practice daily** - Work through examples until automatic
2. **Understand "why"** - Don't just memorize, understand the logic
3. **Use visual models** - Draw fractions, visualize conversions
4. **Memorize common conversions** - Feet/inches, metric, percentages
5. **Practice mental math** - Build speed and confidence
6. **Work through word problems** - Apply fundamentals to real scenarios

---

📚 Next: Apply these fundamentals in Arithmetic Reasoning Study Guide

🔙 Back to Index